System and Devices (SYS3)

Lecture Review – Week 4

Dr Pengcheng Liu

Week 4 Module
Overview and Updates

Module Overview and Updates

- **Week 4**:

Lecture: This week, our discussion was focused on **Pipeline Hazards**. Lecture 6 and Lecture 7 discussed three major types of pipeline hazards: structural hazard, data hazard and control hazard.

Lab: In Week 4, we continued work from Week 3, to complete the development of an algorithm to **resolve structural hazards**, and test it. This week a help sheet is released to allow students to follow a suggested approach for those who have not yet worked out an idea for a solution.

Exercises/Quizzes: Solutions for Exercise 1 will be release on 06/03/2026 (Week 4 Friday). You are expected to complete Exercise 1 by this date. Exercise 1 will be discussed in the Lecture Review session on 12/03/2026 (Week 5 Thursday).

Module Overview and Updates

- **Next Week (Week 5)**:

Lecture: Next week, our discussion will be focused on some **compiler techniques** in Lecture 8. Also, we will be having our second Tutorial Lecture on Pipeline Hazards.

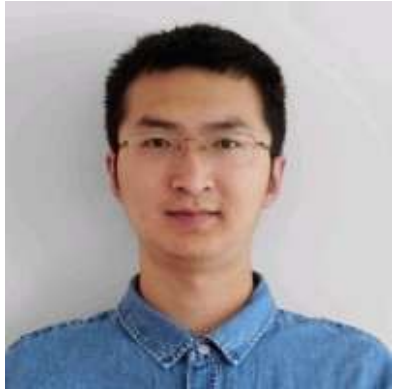
Lab: In Week 5, we will be continue looking at structural hazard. Structural Hazards occur for several reasons, one being the read and write events involving memory, due to the limited number of memory chips, busses, or read and write ports available. This can vary widely depending upon the processor and computer system design. In this practical we will investigate the nature of memory related hazards and attempt to provide suitable solutions.

Exercises/Quizzes: Exercise 2 will be release on 13/03/2026 (Week 5 Friday). Covering Control Hazards, Branch Prediction, Static and Dynamic Scheduling, Superscalar Processors. Please try to complete Exercise 2 by 27/03/2026 (Week 7 Friday). We will use the Lecture Review session on 16/04/2026 (Week 8 Thursday) for discussions if there are any questions on it.

Week 4 Lecture Review

Guest Lecture

Speaker: Dr Yan Wen



Title: Advanced Computer Systems for AI and Computer Vision: From Parallel Architectures to Cloud Deployments

Location: in person at PZA/103 Lecture Theatre, Campus East - Piazza Building

Time: Thursday, 5 March·9:00 – 11:00am

Biography: Dr Yan Wen is an experienced AI scientist with a rich background in computer vision, medical imaging, and data science. With a PhD in Computer Science, Yan has developed advanced algorithms for healthcare applications in lesion analysis, focusing on the challenges of small objects in images using AI. This research has been published in well-known journals/conferences such as ICCV, MIDL, IEEE Access, contributing to the field of computer vision and medical imaging significantly. Currently, as a Data Scientist at the University of Lincoln, Yan leads the development of a medical data platform powered by machine learning, funded by Innovate UK.

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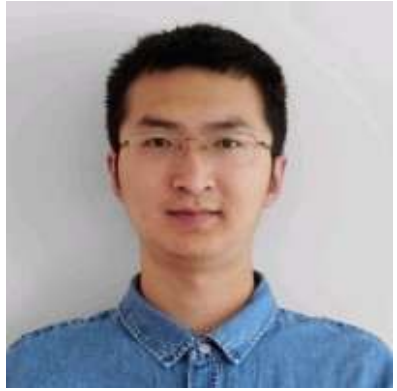

Title: Advanced Computer Systems for AI and Computer Vision: From Parallel Architectures to Cloud Deployments

Abstract: This lecture shows how modern computing hardware helps build and run large-scale computer vision and AI applications. We will start by **comparing CPUs and GPUs**, and explain why doing many operations at once (parallelism) is key for tasks like **matrix multiplication and convolution** in deep learning. Next, we will explore parallel programming by demonstrating a computer vision algorithm on both the CPU and GPU using **PyTorch**, highlighting performance comparisons. Then, we will look at how **AI runs on the cloud at large scale**, using tools like **Kubernetes**, container orchestration, and serverless computing to manage resources. We will also discuss **edge and embedded devices**, explaining how small systems-on-a-chip balance performance and power in tight, real-time conditions. Finally, we will see a case study of a complete **web application** powered by AI and cloud computing. From this lecture, students will understand how to connect these hardware ideas with real-world AI system design, whether running everything on-site or in the cloud.

Thanks

See You Next Week !